



Non-Standard Accent TTS Support via Large Multi-Accent Frontend Pronunciation Knowledge Transfer

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Abstract

Mainstream text-to-speech (TTS) applications rarely offer non-standard accented voice options, perhaps in part due to practical challenges associated with the acquisition of prerequisite training data. Building on prior research, this work demonstrates that a large multi-accent neural frontend model can reduce pronunciation training data requirements by 95% for robust performance in low-resource accents. We further show that pronunciation knowledge transfer is weakly influenced by accent similarity, quantified using a Levenshtein distance-based metric. The large multi-accent paradigm thus emerges as an effective strategy for improving non-standard accent voice-building accessibility, provided source accents are selected to maximize similarity with target accents where possible.

Index Terms: TTS, multi-accent, low-resource, linguistic frontend, transfer learning

1. Introduction

Written text can correspond to strikingly different pronunciations among speakers of a common language. Commercial text-to-speech (TTS) applications, however, generally model only those pronunciations associated with a small subset of highly *standard* accents. This lack of *non-standard* accent representation might lead some to hesitate to interact with voice technologies altogether, because they fail to deliver natural experiences, and might exacerbate negative attitudes towards speakers of such varieties [1, 2].

While a lack of non-standard TTS accent options might cause material representational harm, building systems to accommodate them is non-trivial. High-quality, state-of-the-art (SOTA) speech synthesis typically requires copious data, but the socio-pragmatic status of non-standard accents makes them necessarily *low-resource*. Previous work shows the use of a distinct neural *frontend* module reduces the amount of audio data needed to produce adequate pronunciation performance compared to fully end-to-end systems, by maximizing lexical coverage in its associated pronunciation datasets [3]. However, this strategy relies instead on the availability of large hand-transcribed pronunciation datasets, which are difficult to acquire for many non-standard accents because there are often few, if any, expert users of the relevant language varieties able to produce those transcriptions.

Sun and Richmond [4] present a way to reduce the volume of pronunciation data needed to train neural frontends for novel accents, given existing data associated with other accents. Their “multi-accent” strategy lays the groundwork for keeping both text and audio data to a minimum, which is essential to facilitate greater non-standard accent voice-building accessibility.

Sun and Richmond [4] demonstrate that a multi-accent

frontend can compensate for the sparsity of low-resource accent pronunciation data with strong lexical coverage in an alternate one. They show that by “augmenting” the training data of one source accent with extra sentences, the correct pronunciations for those sentences in a separate target accent (for which the sentences are unseen) are “transferred” at impressive rates. Specifically, the researchers report as few as 20k target accent training sentences (roughly 25.5k word types) yield pronunciation transfer success commensurate with the accuracy of a baseline model trained on 206k target accent sentences.

Unfortunately, 20k lines of hand-transcribed pronunciation data is still far too many for low-resource voice-building contexts. Reducing the number to one a single developer could manage, for example, would be a great next step. Additionally, Sun and Richmond’s reported effect is supported by only one configuration of source/target accent pair whose degree of similarity was not quantified. It is possible, however, that the multi-accent model performs better when source/target accent sequences have a greater overlap of phonetic characters, which would be consistent with findings associated with multilingual large language models [5, 6, 7, 8]. This uncertainty prompts the need for an extension of Sun and Richmond’s experiments with new pairs whose degrees of similarity are quantified objectively.

The present work thus expands on Sun and Richmond’s [4] efforts to reduce the training data needed to support novel-accent TTS development via multi-accent pronunciation knowledge transfer. We present two novel contributions. First, we find ceiling-level word transcription accuracy rates can be achieved via a new record-low number of training data sentences—a 95% decrease from Sun and Richmond’s demonstrated reduction—by increasing the number of accents in the multi-accent model. Second, we find that the observed pronunciation knowledge transfer effect that enables a reduction in training data volume is indeed weakly mediated by the similarity of source and target accent pairs, as quantified via Levenshtein distance. The large multi-accent paradigm therefore emerges as a robust technique for adding non-standard accent frontend coverage, provided developers seek to maximize pronunciation knowledge transfer by using a maximally similar source accent.

2. Model Architecture

All models presented in this work are neural Sequence-to-Sequence (Seq2Seq) frontend models whose structures and hyperparameters adhere to Sun and Richmond’s [4] design. For a full account, the reader is referred to their work.

The frontend model translates a text sentence character sequence $x_{1:T} = [x_1, x_2, \dots, x_T]$ to a linguistic specification character sequence $y_{1:L} = [y_1, y_2, \dots, y_L]$ (where T and L are lengths) via a bidirectional LSTM encoder and unidirectional LSTM de-

coder augmented with an attention mechanism. The attention mechanism is a monotonic Gaussian mixture model (GMM) [9]. Each model is trained on a single NVIDIA Tesla V100-SXM2-16GB GPU accelerator until convergence, determined according to validation loss (typically 2–3 days). All reported evaluation scores are the average of runs of inference at three post-convergence saved parameter checkpoints.

3. Data

3.1. Source Text & Target Linguistic Specification

Training and validation datasets are derived from the Libri-TTS corpus and test datasets are from the HiFi-TTS corpus [10, 11]. Pronunciation training data comprise 14 sets of 199,612 sentences of paired English text/linguistic specification sequences. Validation and test data comprise 14 sets of 2181 and 4431 sentences of paired sequences, respectively.

Target linguistic specifications are “bootstrapped” from the Festival (pipeline-based) frontend [12]. A sample sentence is given in Table 1. The “accent-independent” phoneset system known as *Unilex* is utilized to enable the simultaneous modeling of many unique accents with a single shared phone superset.¹

Input (Text)	Output (Linguistic Specification)
AT ABOUT THE HEIGHT OF A MAN’S SHOULDERS ABOVE THE STREET.	1 a t + 0 @ - 1 b ow t + 0 dh @ + 1 h ai t + 1 uh v + 1 ei + 1 m eh n z + 1 sh ou lw - 0 d @r r z _B 0 @ - 1 b uh v + 0 dh @ + 1 s t r ii t _BB

Table 1: Example of Libri-TTS input text and corresponding ground-truth frontend output for the General American (GAM) accent. The output includes phone symbols and labels for lexical stress (0, 1, 2), word boundaries (+), syllable breaks (-), and prosodic boundaries (.B, .BB), consistent with Festival conventions.

3.2. Accents & Bootstrapping Procedure

The creators of *Unilex* maintain a rule-system for mapping baseline phone sequences to accent-specific ones, for 14 distinct English accents. All 14 accents, displayed in Table 2, are included in this study.

Festival’s frontend pipeline involves a lexicon look-up and statistical grapheme-to-phoneme (G2P) back-off. To minimize error associated with ground-truth pronunciation data, we only bootstrap sentences for which each word is contained in Festival’s lexicon, consistent with Sun and Richmond [4]. We also repeat their exclusion of text normalization from the definition of the present frontend task. Frontend pronunciation data is thus attained by removing “out-of-vocabulary” (OOV) tokens from the Libri-TTS and HiFi-TTS corpora, and then passing them, line by line, through 14 unique configurations of Festival’s frontend, each initialized with a different accent-specific translation of the master lexicon (made possible by the *Unilex* accent translation rules).

3.3. Accent Similarity

We define accent similarity as a score based on Levenshtein’s minimum edit distance [13], adapted to compare lists of strings

Nation	Region	Accent Name	Accent Code
Australia	Australia	General Australian	GAU
Ireland	County Clare	County Clare	CCL1
NZ	New Zealand	General New Zealand	GNZ
UK	Northern England	Leeds	LDS
UK	Northern England	LDS—no-/h/-drop	LDS1
UK	Scotland	Edinburgh	EDI
UK	Scotland	Aberdeen	ABD1
UK	Southern England	Received Pronunciation	RPX
UK	Wales	Abercrave	ABC
UK	Wales	Cardiff	CDF
USA	Northeast	New York City	NYC
USA	Northeast	NYC—non-rhotic	NYC1
USA	North E./W. Coast	General American	GAM
USA	South	South Carolina	SCA

Table 2: The 14 *Unilex*-supported English accents examined in this work, presented alphabetically by nation then region.

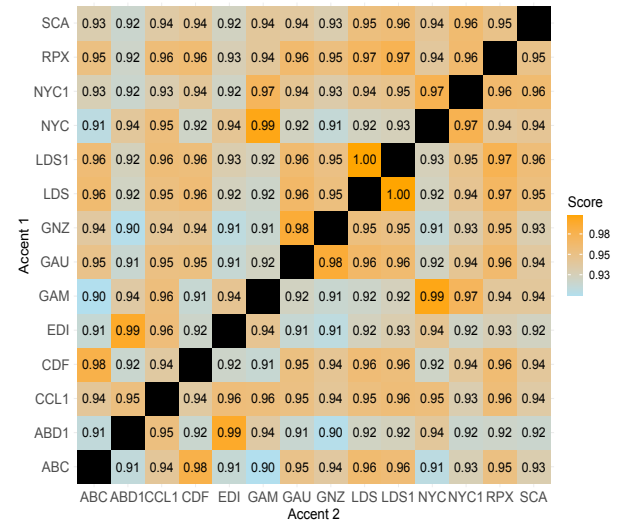


Figure 1: Heatmap of all *Unilex*-supported accent pair similarity scores, rounded to the nearest hundredth. Used for operational definitions of low, moderate, and high similarity accent pairs.

(because a single *Unilex* phone can be one or multiple characters). For each unique word present in the full Libri-TTS source data, the Levenshtein distance between its phonetized forms in two target pronunciation datasets is calculated. This distance is subtracted from the sum of the lengths of both phonetized words and then divided by that sum, yielding a normalized “Levenshtein ratio” for each word.² A final similarity score is thus obtained by averaging the normalized Levenshtein ratios across all unique words in the datasets. Scores range from 0 (no similarity) to 1 (identical).

Accent distance is computed for every combination of accents examined in this study (91 pairs). All (rounded) similarity scores are presented in Figure 1; the median and range of each accent’s score with all others are depicted in Figure 2.

¹<https://www.cstr.ed.ac.uk/projects/unisyn/>

²<https://github.com/closetothe/levenshtein/blob/master/levenshtein.py>

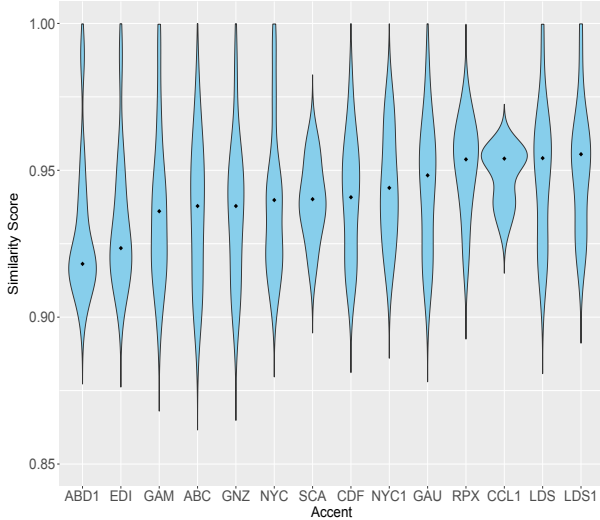


Figure 2: Violin plot depicting similarity score distribution for each accent compared to all other accents, sorted in ascending order by median values (black dots).

4. Experiments

4.1. Large Multi-Accent Baseline

As a preliminary validity check, we first confirm the incorporation of 14 accents into one multi-accent model—a large expansion from Sun and Richmond’s [4] tri-accent model—does not degrade performance compared to two baseline uni-accent models. EDI and LDS1 are designated as test accents for this experiment.

Test Accent	Uni-Accent	Multi-Accent
EDI	96.75	96.33
LDS1	95.71	96.30

(a) Word boundary & prosodic boundary accuracy (%)

Test Accent	Uni-Accent	Multi-Accent
EDI	99.95	99.89
LDS1	99.71	99.76

(b) Seen word accuracy (%)

Test Accent	Uni-Accent	Multi-Accent
EDI	78.28	78.56
LDS1	83.52	80.70

(c) Unseen word accuracy (%)

Table 3: Accuracy rates for multi-accent model tested on EDI and LDS1 target accents, compared to uni-accent baselines.

EDI and LDS1 word boundary and prosodic boundary label accuracy rates for uni and multi-accent models are presented in Table 3a. All scores appear high and within a small range. The word accuracy rates (which consider phone symbols, lexical stress labels, and syllable break labels), are given in Tables 3b and 3c, for both seen and unseen word tokens associated with all sentences found to contain no boundary errors. The accuracy scores associated with *seen* word tokens confirms that all models *memorize* their training data effectively (all scores are well

above 99%). Lastly, the accuracy scores associated with *unseen* word tokens do not suggest a breakdown in the model’s ability to perform the frontend task when increasing the number of accents modeled simultaneously from 1 to 14. Interestingly, though, unseen word accuracy scores are higher for LDS1 than EDI, for both uni and multi-accent configurations. One possible explanation for this is the LDS1 accent *reduces* vowels far more than EDI (The LDS1 dataset contains nearly three times as many schwas), which could result in “easier” data to learn. Overall, our results show a large number of accents can be successfully covered by a single model.

4.2. Scaling of Pronunciation Knowledge Transfer with Large Multi-Accent Models

Next, we evaluate the performances of multi-accent models trained on increasingly smaller datasets but augmented for one source accent only with a separate target accent’s test data sentences, to determine the smallest size that yields ceiling-level knowledge transfer results. EDI is designated as target accent for all experiments. The augmented source accent is selected such that it and EDI form a high-similarity pair: ABD1 (similarity score = 0.99). In these experimental configurations, target accent training data is sampled randomly without replacement from the Libri-TTS data such that none of those sentences are also used as training data for the other accents (but all other accents share the same training sentences).³ The lack of overlap between target accent and other accent training data replicates Sun and Richmond’s [4] design and reflects the fact that newly acquired data for novel accent development may be unique. The reduced-data experimental models are compared to a ceiling-baseline model in which the full Libri-TTS dataset is used as training data for all 14 accents (but one source is still augmented as in the experimental models), as depicted in Figure 3.

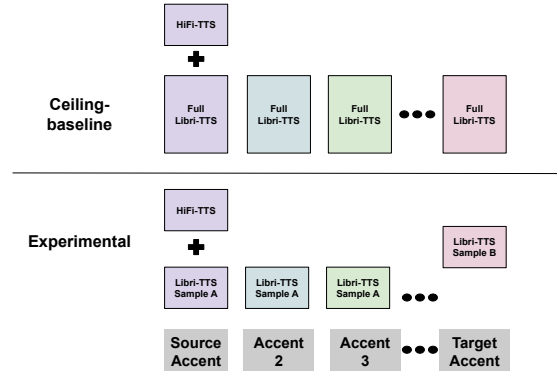


Figure 3: Training configurations for ceiling-baseline and experimental models. Ceiling-baseline models use full Libri-TTS dataset for all accents. Experimental models use two random sub-samples of Libri-TTS dataset so that target accent data has no sentences in common with other accent data. Sub-sample sizes are manipulated as an independent variable. All models augment source accent data with target accent’s test data.

This scaling methodology departs from Sun and Richmond’s design in one key respect: the previous experiments

³In rare cases where random selection fails to facilitate full coverage of the Unix phoneset (e.g., by excluding /x/), sentences containing the missing phones are added manually to ensure full coverage.

utilized a constant, much larger dataset size for non-target accents (excluding augmentation data) in all configurations. In the present design, however, the number of target and non-target (non-augmentation) training sentences is scaled down jointly to limit the scale of imbalance that would be introduced if only the target accent’s data were scaled down, given such an extensive increase in intermediate accent data. Namely, if only 1k sentences of sequences are seen for the target accent, for example, but a much larger number of sentences are seen for some 13 others accents, overfitting of the non-target data as a consequence of its over-representation seems likely [14]. Moreover, we confirm this assumption in a post-hoc replication of our record-low model configuration with an alternate version in which *only* the test accent is scaled down. This configuration indeed fails to facilitate a reproduction of our results.

Results associated with the ceiling-baseline, the lowest experimental data size to yield ceiling-level performance, and the first experimental data size to fail are displayed in Table 4. The new lowest size that achieves ceiling-level accuracy—1k sentences—is only about 0.5% of the size of the ceiling-baseline’s training dataset, and signifies a large decrease compared to the previous SOTA of 20k (a 95% reduction). The scaling experiment results thus suggest that the substitution of a tri-accent model with a large multi-accent model reduces data requirements substantially by strengthening the effect of pronunciation knowledge transfer. We cautiously speculate that the additional intermediate accent examples provide a bridge that enable the model to better “pick out” the key patterns by which an accent departs from invariant features of English phonology.

Model Configuration	# Word Types	Accuracy (%)
Ceiling-Baseline	47,055	88.32
Experimental-1k	5,870	89.21
Experimental-500	3,870	x

Table 4: *High-similarity (Source=ABD1, Target=EDI) scaling of unseen word accuracy scores. Ceiling-baseline trained on 199,612 sentences; experimental models trained on 1,000 and 500 sentences respectively. X indicates complete model failure.*

4.3. Effect of Accent Pair Similarity on Scaling of Pronunciation Knowledge Transfer

Having determined that 1k training sentences is the minimum that maintains word accuracy levels consistent with the ceiling-baseline for a high-similarity accent pair (ABD1/EDI; score = 0.99), the ceiling-baseline, experimental-1k, and experimental-500 model configurations are repeated for additional moderate similarity (GAM/EDI; score = 0.94) and low-similarity (GNZ/EDI; score = 0.91) accent pairs. The accuracy scores associated with the three accent pairs for each model configuration are given in Table 5.

Accent Pair	Similarity	Accuracy (%) Baseline / 1k / 500
ABD1/EDI	High	88.32 / 89.21 / x
GAM/EDI	Moderate	87.24 / 87.28 / x
GNZ/EDI	Low	89.72 / 87.44 / x

Table 5: *Unseen word accuracy scores for ceiling-baseline, experimental-1k, and experimental-500 models across accent pairs of varying similarity. Bold indicates first model failure or significant performance drop ($p < 0.05$) relative to baseline.*

While the moderate-similarity accent pair models appear to behave similarly to the high-similarity pair ones, there is evidence of a reduction in the strength of pronunciation knowledge transfer upon substituting these with low-similarity pair models. Specifically, the lowest successful training data size of 1k sentences illustrated with {ABD1/EDI} does not replicate with {GNZ/EDI}—a significant decrease in unseen word accuracy rate occurs at this threshold compared to its ceiling-baseline ($p < 0.05$). This finding suggests the similarity of source and target accent is indeed a relevant factor to consider when developing multi-accent strategies for low-resource accent frontend support, although even moderate similarity appears to facilitate success.

5. Discussion

The evidence presented in this work is limited by two design choices. First, the finding that accent similarity affects pronunciation knowledge transfer is based on only one example of each similarity type (low, moderate, high), whereas replication with additional pairs would provide stronger evidence. Second, the use of bootstrapped pronunciation data might limit the external validity of these experiments, as the “real-world” annotations it seeks to simulate may be error-prone to some degree. Future research should consider data noising techniques to more closely model human data [15]. Nevertheless, we find compelling preliminary evidence that the use of a large multi-accent frontend model enables strong linguistic specification transfer, provided the similarity between source and target accents is maximized.

A compelling next step would be to develop a single pre-trained multi-accent model that could be *fine-tuned* for any new low-resource accent desired. While additional experimentation would be needed to establish the optimal model calibration that avoids so-called catastrophic forgetting, or over-application of the target accent’s phonology at the cost of knowledge contributed by the other accents’ pronunciations, such work could lead to even further advances [16].

6. Conclusion

In this work, we show a large neural multi-accent frontend model can facilitate strong linguistic specification transfer with a record low number of novel accent training examples—1k for high and moderate-similarity pair configurations. This reduction in training data volume needed to produce robust frontend behavior suggests the large neural multi-accent frontend can improve non-standard accent voice-building accessibility by substantially reducing its seemingly insurmountable data sourcing obstacle. The failure to replicate this new record with a low-similarity accent pair configuration, on the other hand, indicates that careful planning should go into the selection of the source accent to maximize pronunciation knowledge transfer success where possible.

Finally, while we believe the improvement of non-standard accent representation is important as we strive for greater inclusivity in science and technology, a word of caution is raised. It is possible that the availability of some non-standard accents in certain mainstream TTS applications could lead to situations regarded as appropriation and/or mockery [17]. System developers must therefore carefully consider the ethical implications of their particular application before seeking to introduce new accents, and establish safeguards to help mitigate potential harms [18].

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